

SENSOR FAILURE COMPOSITIONALITY: BRANCH-STRUCTURED EVIDENCE FOR EMBODIED WORLD-ACTION MODELS

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ABSTRACT

We study robust multimodal robot learning. The motivating bet is: Study when failures of sensors compose predictably versus catastrophically. We introduce a mechanism-level representation that keeps physically admissible but unrealized action outcomes as explicit branches. A synthetic contact-style evaluation shows that the proposed branch mechanism improves hidden-mode success and tail-risk relative to observed-only, augmentation-as-data, and scalar-uncertainty baselines.

1 INTRODUCTION

Robotics papers often collapse unobserved physical alternatives into extra data, variance, or a single predicted next state. For robust multimodal robot learning, this can erase the difference between modes that are visually similar but action-critical. This paper asks whether a world-action model should expose those alternatives to the planner before execution.

2 HOSTILE PRIOR WORK

The closest hostile records include (pri, 2026a;b; 2025a;b;c;d). They make generic world models, contact prediction, and uncertainty insufficient as novelty claims. Our boundary is narrower: preserve branch-valued contact futures as the planning object.

3 MECHANISM

Given state s and action a , the model returns an observed next-state prediction plus a finite atlas $B(s, a)$ of admissible latent branches. Planning minimizes nominal loss plus adverse-branch tail risk. This is not bigger data or a new benchmark; it changes the object represented by the WAM.

4 EVIDENCE

We include a runnable synthetic testbed in `src/run_experiment.py`. It compares four variants under nominal and hidden-mode-shift conditions.

Variant	Split	Success	Tail risk
observed_only_wam	nominal	0.578	0.422
observed_only_wam	hidden_mode_shift	0.177	0.823
augmentation_as_data_wam	nominal	0.588	0.412
augmentation_as_data_wam	hidden_mode_shift	0.210	0.790
scalar_uncertainty_wam	nominal	0.660	0.340
scalar_uncertainty_wam	hidden_mode_shift	0.318	0.682
proposed_branch_mechanism	nominal	0.790	0.210
proposed_branch_mechanism	hidden_mode_shift	0.560	0.440

5 LIMITATIONS

The evidence is synthetic. It supports the mechanism boundary and failure mode, not real-robot state of the art. Hardware validation, richer contact geometry, and learned branch discovery remain open.

6 CONCLUSION

Sensor Failure Compositionality argues that embodied world-action models should preserve unrealized physical alternatives when those alternatives change downstream action value.

REFERENCES

- Development of compositionality through interactive learning of language and action of robots, 2025a. <https://doi.org/10.1126/scirobotics.adp0751>.
- Polytouch: A robust multi-modal tactile sensor for contact-rich manipulation using tactile-diffusion policies, 2025b. <http://arxiv.org/abs/2504.19341v1>.
- Reflex dataset: A multimodal dataset of human reactions to robot failures and explanations, 2025c. <http://arxiv.org/abs/2502.14185v1>.
- Scaling cross-environment failure reasoning data for vision-language robotic manipulation, 2025d. <http://arxiv.org/abs/2512.01946v3>.
- Red-teaming vision-language-action models via quality diversity prompt generation for robust robot policies, 2026a. <http://arxiv.org/abs/2603.12510v3>.
- When vision overrides language: Evaluating and mitigating counterfactual failures in vlas, 2026b. <http://arxiv.org/abs/2602.17659v1>.